

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – Short Answer

Course Code: ITA03 CO Assessed: CO1 Weightage: 30 % of CIA CO1 Statement: Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems. Student Name: Reddy Raghuvardhan Section / Batch: CSE(A1)	Course Title: Mobile computing Assessment: Assessment Tool 2 – Short Answer Total Marks: 100 Marks Reg. No.: 1923 T2248 Date: 23/07/2026
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Code of Conduct

I, Reddy Raghuvardhan (Name / Reg No) 1923T2248 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: R. Raghuvardhan

Instructions

- This test contains 50 questions. Total: 100 Marks.
- Duration: 1 hour.

90
100

A telcom operator wins a spectrum auction for 45 MHz. company policy sets aside 25 kHz for every simplex channel. The planning team...

Total spectrum = 45 MHz. Full-duplex channel needs 2 simplex channels = $2 \times 25 = 50 \text{ kHz}$. $45,000 / 50 = 900$
 Answer = 900

2 sol

Solution: channels per cell = total channels / N = $560 / 7 = 80$

3 sol

Total voice channels = carriers x slots / carrier = $100 \times 8 = 800$

4 Ans

power ratio = $80 / 4 = 20$ Ratio in dB = $10 \times \log_{10}(20) = 13.01 \text{ dB} \approx 13 \text{ dB}$

5 Ans

C/I (dB) = Signal (dBm) - interference (dBm) = $60 - (-42) = 102 \text{ dB}$

6

total channels = $42,000 \text{ kHz} / 300 \text{ kHz} = 140$. channels per cell = $140 / 7 = 20$

7

Received power = $1/d^n$. doubling distance with $n=4$ reduces power by $2^4 = 16 \times$ new power = $64 / 16 = 4$

8

with $n=2$ (inverse square law). doubling distance with power increased by $2^2 = 4 \times$ to maintain same edge SNR. Answer = $4 \times$

(9) Total channels in cluster = cluster's size $\times$ channels/cell
 $= 4 \times 15 = 60$

(10) $v = 160 \text{ km/hr} = 27.78 \text{ m/s}$. Doppler shift $= v \times f / c = 27.78 \times 900 \times 10^6 / 3 \times 10^8 = 83.3 \text{ Hz} \approx 83 \text{ Hz}$

(11) slot duration = Frame time / slots $= 5 \text{ ms} / 8 = 0.625 \text{ ms}$

(12) carriers = total spectrum / spacing $= 24,000 \text{ KHz} / 200 \text{ KHz} = 120$

(13) Tx power in dBm $= 10 \log_{10} (8000 \text{ mW}) = 39.03 \text{ dBm}$ Received power $= 39.03 - 33 = 6.03 \text{ dBm} = 6 \text{ dBm}$

(14) Using the log-distance path loss model referenced to 1m (consistent with -120 dB at 2 km for a nearby frequency band). scaling the distance at 2.5 km gives an approximate free-space path loss of -138 dB .

(15) Total capacity = calls/cell $\times$ cells $= 250 \times 6 = 1500$

Active calls supported = arrival rate $\times$ avg duration =
 $4000 \text{ calls/sec} \times 180 \text{ sec} = 720,000$

(17) Total bandwidth = channels $\times$ bandwidth/channel = $150 \times 25 \text{ KHz} = 3750 \text{ KHz} = 3.75 \text{ MHz}$.

(18) Number of guard bands between 30 bands = 29. wasted bandwidth = $29 \times 40 \text{ KHz} = 1160 \text{ KHz} = 1.16 \text{ MHz}$.

(19) speed = $90 \text{ km/hr} = 25 \text{ m/s}$. cell diameter = $2 \times 1.5 \text{ km} = 3000 \text{ m}$.
Time per cell = $3000 / 25 = 120 \text{ s}$. handoff/hour = $3600 / 120 = 30$.

(20) Total users = regional centers $\times$ base stations/center
 $\times$ users/base station = $8 \times 40 \times 250 = 80,000$.

(21) Approximate cell area = $R^2 = (0.4 \text{ km})^2 = 0.16 \text{ km}^2$. Number of cells = $400 / 0.16 = 2500$

(22) channels = Total spectrum / channel width = $6000 \text{ KHz} / 30 \text{ KHz} = 200$.

(23) Total call-minutes = $1200 \times 4 \times 2.5 = 12,000 \text{ min/hr}$.
Traffic in Erlangs = $12,000 / 60 = 200$

24) $\text{users} = \text{total bandwidth} / \text{bandwidth per user} = 40,000 \text{ KHz} / 400 \text{ KHz} = 100$

25) $\text{total capacity} = \text{users/cell} \times \text{cells} = 600 \times 1200 = 720,000$

26) $\text{paging operations} = \text{total area cells} / \text{cells per operator} = 200 / 4 = 50$

27) $\text{offered traffic} = \text{arrival rate} \times \text{avg duration} = 401,600 \times 3 = 1,204,800 \text{ Erlangs}$
 $\text{Excess over capacity} = 1,204,800 - 1,000,000 = 204,800$
 $\text{Fraction blocked} = 204,800 / 1,204,800 = 0.17$

28) $\text{Roaming users} = 15\% \times 600,000 = 90,000$

29) $\text{utilization} = \text{peak traffic} / \text{processing rate} = 90,000 / 120,000 = 0.75 = 75\%$

$\text{Total channels} = \text{Bandwidth} / \text{channel width} = 48,000 / 4 = 12,000$
 $\text{channels per cell} = 12,000 / 4 = 3,000$

$$\text{channels} = \text{total spectrum} / \text{channel width} = 30,000 / 150 = 200$$

$$\text{Successful handoffs} = 97\% \times 2500 = 2475$$

(32)

Ans:-

$$\text{TX power in dBm} = 10 \log_{10}(12,000 \text{ mW}) = 40.79 \text{ dBm}$$

(33)

Ans:-

$$\text{Received power} = 40.79 - 45 = -4.21 \text{ dBm}$$

(34)

Ans:-

$$\text{Total capacity} = \text{users/cell} \times \text{cells} = 900 \times 7 = 6300$$

(35)

Ans:-

$$\text{slot duration} = \text{frame time} / \text{slots} = 4.8 \text{ ms} / 8 = 0.6 \text{ ms}$$

(36)

Ans:-

$$150 \text{ km/hr} = 41.67 \text{ m/s}, \text{ Doppler shift} = v \times f/c = 41.67 \times 1.8 \times 10^9 / 3 \times 10^8 = 250 \text{ Hz}$$

(37)

Ans:-

$$\text{offered traffic} = 70,000 \times 4 = 280,000 \text{ Erlangs}$$

$$280,000 - 250,000 = 30,000$$

$$\text{Fraction blocked} = 30,000 / 280,000 = 0.11$$

(38)

Ans:-

$$\text{Total users in cluster} = \text{users/cell} \times \text{cells in cluster}$$

$$= 600 \times 9 = 5400$$

39) calls dropped/sec = $0.4\% \times 60,000 = 240$.

Ans

40) path loss (dB) = TX power (dBm) - RX power (dBm) = $43 - (-87) = 130 \text{ dB}$.

Ans

41) Total coverage = coverage/station $\times$ number of stations = $2.5 \times 150 = 375 \text{ km}^2$.

Ans

42) Total channels = $48,000 / 200 = 240$ channels per cell
 Ans = $240 / 6 = 40$.

43) paging cycles = Total users / users per cycle = $150,000 / 6000$
 Ans = 25 .

44) Handoffs per cell = call duration / handoff intervals
 Ans = $9 / 3 = 3$.

45) SNR (dB) = signal (dBm) - noise (dBm) = $-85 - (-115) = 30 \text{ dB}$.

Ans

46) Minimum bandwidth = users $\times$ rate / users = $250 \times 150 \text{ kbps}$
 Ans = $37,500 \text{ kbps} = 37.5 \text{ Mbps}$.

Ans

47) Total call-minutes/hr = users $\times$ calls/hr $\times$ avg duration
 = $1500 \times 4 \times 2.5 = 15,000 \text{ min}$. Traffic in Erlangs = $15,000$

48) Total bandwidth = channels $\times$ bandwidth/channel = 3 MHz .
 Total throughput = connections $\times$ rate/connection = 960 Mbps .
 49) Area per tower = $\pi R^2 = \pi \times 4^2 = 50.27 \text{ km}^2 = 300 / 50.27$
 50) = $6.76 = 7$.